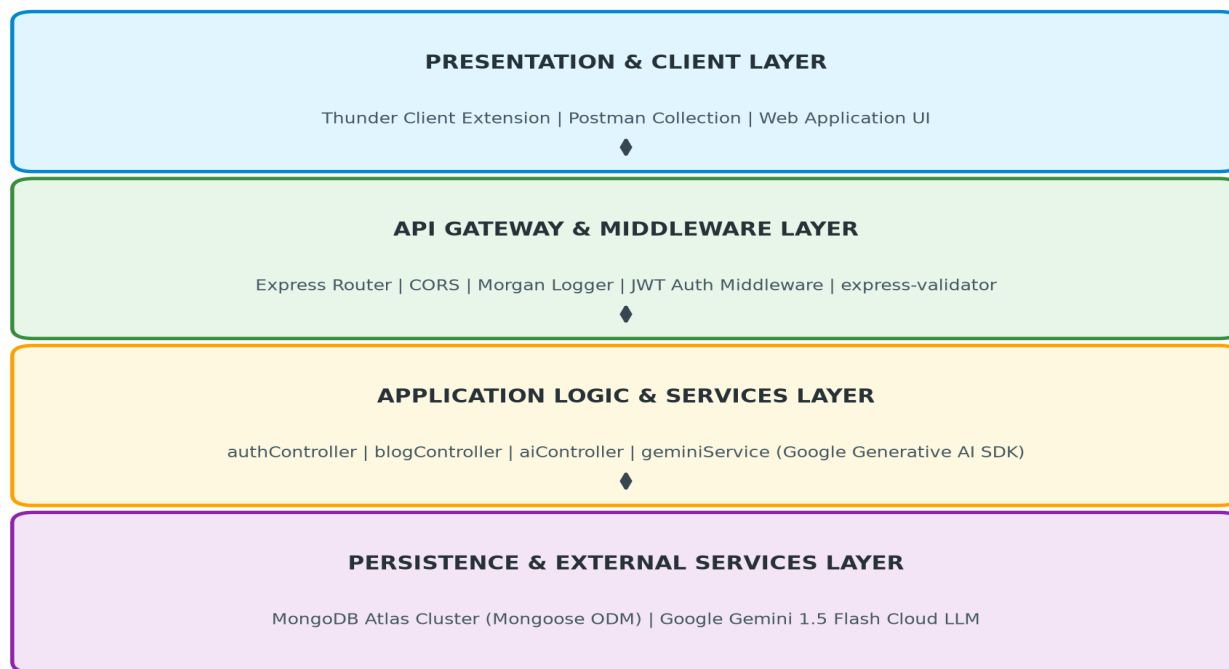


Solution Architecture Diagram

Date:	3 October 2026	Team ID:	UNM1611SHARVESHARAM-SWTID-2026-8110
Project Name:	AI BlogNest API	Maximum Marks:	5 Marks

Solution Architecture Blueprint:

AI BlogNest API is designed using a multi-layered, decoupled architecture separating presentation, gateway routing, business logic, security middleware, and external data services.



Architectural Highlights:

- **Stateless REST Design:** Every request is self-contained with JWT Bearer authentication.
- **Decoupled AI Engine:** geminiService isolates LLM communication from HTTP route handlers.
- **Centralized Resilience:** Single error middleware catches synchronous and asynchronous failures.

Solution Architecture — Component Description Table

Architectural Layer Component Specifications:

Component Name	Layer	Role & Responsibilities	Technologies Used
Presentation Client	Presentation	Sends REST HTTP JSON requests (Register, Login, Generate, CRUD). Manages test collections and bearer authorization tokens.	Thunder Client, Postman v2.1, Browser HTTP Client
API Gateway & Server	Gateway	Initializes Express application, mounts CORS policy, JSON body parser, Morgan HTTP logging, and environment variable configuration.	Node.js, Express.js (app.js, server.js), cors, morgan, dotenv
Security & Validation	Middleware	Enforces Bearer token extraction and verification; rejects unauthenticated requests with 401; validates request schemas with 422 errors.	jsonwebtoken, bcryptjs, express-validator, authMiddleware.js
Controllers & Services	Application Logic	Implements business logic for auth, blog operations, prompt building, regex text cleaning, and domain category classification.	authController.js, blogController.js, aiController.js, geminiService.js
External AI Service	External AI	Executes generative LLM queries to produce 100-150 word blog posts and concise executive text summaries.	Google Gemini API, gemini-1.5-flash, @google/generative-ai
Data Storage Layer	Persistence	Stores and indexes Users and Blogs collections. Enforces schema constraints, default timestamps, and author ObjectId references.	MongoDB Atlas, Mongoose ODM (User.js, Blog.js, db.js)

Security Boundaries & Fault Tolerance:

- **Credential Isolation:** MongoDB connection strings and Google Gemini API keys reside exclusively in .env and are never hardcoded or committed to git.
- **Resource Isolation:** A user cannot modify or delete articles authored by other users (returns 403 Forbidden).
- **Graceful Fallback:** If GEMINI_MODEL environment is misconfigured, geminiService automatically falls back to gemini-1.5-flash.